

Physical Education Knowledge and Self-Reported Behavioral Outcomes

James Ordway

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Abstract

Physical education (PE) programs play an important role in promoting physical health among youth, yet curriculum design is often guided by tradition and personal preferences rather than empirical evidence. This study examines the relationship between standards-based PE knowledge and self-reported physical behavior outcomes among high school students, using a cross-sectional dataset collected from 1,504 students across 9 Indiana secondary schools during the 2018-2019 academic year. Knowledge was evaluated using a 39-item assessment based on the three knowledge-based SHAPE National Standards (Standards 2, 3 and 4), and physical activity was measured using three YRBS-derived outcomes: PA Minutes per Day, PA Days per Week, and Fitness Cohort membership. Three regression models were fit for each outcome, with and without demographic covariates. Across all three outcome measures, higher PE knowledge scores were associated with higher levels of physical activity, even after controlling for demographic variables. Standard 2, which corresponds to general fitness knowledge, was the most consistent predictor across models. Effect sizes were small and model fit was poor, suggesting that PE knowledge only explains a small portion of the variance in physical activity behavior. These findings serve as a proof-of-concept for the potential of standards-based surveillance data to inform evidence-based PE curriculum design, and to motivate larger-scale, more representative research.

Introduction

2.1 SHAPE Standards Framework

Physical education (PE) programs play an important role in promoting the physical health and activity levels of children across the United States. To provide structure for this mission, the Society of Health and Physical Educators (SHAPE) established four national standards to guide educators in their provision of holistic PE programming. These standards organize physical literacy into four categories: (1) motor skill development, (2) the application of movement and fitness knowledge, (3) the development of social skills through movement, and (4) the development of personal identity through movement (SHAPE America – Society of Health and Physical Educators 2024).

2.2 The Evidence Gap

Despite the existence of these standards, current PE programming relies heavily on historical traditions, anecdotal assumptions, and personal preferences of individual instructors rather than large-scale empirical evidence (Richards et al. 2014). This lack of evidence contributes to the inconsistency across programs, and may limit their effectiveness in promoting physical activity in students. It is widely acknowledged that there are too many standards and that programs should be prioritized based on teacher philosophy, program foci, and student/community interests (Lund and Tannehill 2015). Meanwhile, national surveillance data continue to highlight a concerning trend in sedentary behavior among children (Pfledderer et al. 2025; Tremblay et al. 2011). Large-scale, standards-based research has the potential to identify which aspects of PE are associated with positive behavioral outcomes, and inform curriculum design as educators work to improve the physical health of their students.

2.3 Purpose of Study

This study serves as a proof-of-concept for the potential of standards-based surveillance data in providing an empirical framework for PE programming. Leveraging a unique dataset that contains validated SHAPE standards-based assessment items and Youth Risk Behavior Surveillance (YRBS) self-report measures, we examine how PE knowledge standards relate to self-reported physical activity and fitness levels. Findings from this pilot are intended to motivate larger-scale, more representative research that can ultimately support data-driven curriculum design.

Methods

3.1 Study Design and Setting

This study utilized a cross-sectional survey dataset collected from secondary schools in Indiana. The survey was administered to a total of 1,504 students from 19 classrooms across 9 schools during the 2018-2019 school year. Accordingly, the data has a hierarchical structure, with students belonging to classrooms within schools.

3.2 Description of Data Sources

The physical activity behavior data were drawn from the Youth Risk Behavior Survey (YRBS), which is developed and maintained by the CDC. The YRBS is part of the Youth Risk Behavior Surveillance System (YRBSS), the largest public health surveillance system in the United States, designed to monitor health risk behaviors contributing to the leading causes of mortality, morbidity, and social problems among our youth. The survey is self-administered and designed to be completed in approximately one class period (45 minutes) (Underwood et al. 2020).

The dataset includes a 39-item assessment of students' PE knowledge (see Appendix A), as well as their self-reported physical behavior outcomes from the YRBS survey. The knowledge assessment items were originally developed under the 2004 SHAPE National Standards framework, which included 6 standards, and were subsequently mapped onto the three knowledge-based standards (Standards 2, 3 and 4) of the current four-standard framework. Student learning outcomes of the original 6-standard framework were cross-checked with the current 4-standard framework, and assigned to the appropriate new umbrella standard.

3.3 Variable Construction

3.3.1 Independent Variables

Of the 39 assessment items, 21 corresponded to Standard 2 (fitness knowledge), 4 to Standard 3 (social skills), and 14 to Standard 4 (personal identity/values). Raw scores for each cluster were aggregated into percentage-correct variables, which were then normalized (mean=0, sd=1). Standard 1 corresponds to motor skill development, and thus is not appropriately measured by a written assessment.

3.3.2 Dependent Variables

We used three self-reported behavioral outcome metrics derived from the YRBS:

Physical Activity (PA) Minutes per Day: Students were asked to report their average daily minutes of physical activity as belonging to one of four categories. These categories included 0-20 minutes, 20-40 minutes, 40-60 minutes, and 60+ minutes.

Physical Activity (PA) Days per Week: Students reported the number of days per week that they were physically active for at least 60 minutes. Responses ranged from 0 to 7 days.

Fitness Cohort: Students were separated into three “fitness cohorts,” based on both PA minutes per day, as well as the average number of days that they reported training physical strength (0-7). The “High Fitness” cohort was comprised of students meeting the recommendation of 60+ PA minutes per day, as well as 3+ days

of strength training per week. The “Low Fitness” cohort included students that reported 0-20 PA minutes per day and less than 2 days of strength training per week. The “Medium Fitness” category included all students not meeting the High or Low criteria.

3.4 Missing Data

A small number of observations were missing values for dependent variables, including 6 for PA Days per Week, 3 for PA Minutes per Day, and 2 for Strength Days. These observations were excluded from their respective models during analysis.

3.5 Modeling

To examine the relationship between PE test scores and physical activity, three distinct regression techniques were used, one for each dependent variable. All analyses were conducted in R.

Exploratory data analysis revealed that test scores increased significantly from the lowest activity category (0-20 minutes) to the second-lowest (20-40 minutes), but scores plateaued after increasing to higher activity categories. As a result, PA Minutes per Day was dichotomized into a binary variable indicating whether or not a student fell above or below the 20-minute threshold, and a logistic regression was fit to model the relationship between assessment scores and sedentary behavior.

PA Days per week was modeled using a binomial regression, as it can be interpreted as a count of successes (days physically active) out of fixed trials (7 days in a week).

The Fitness Cohort variable was dichotomized into a binary variable due to the relatively low sample size of the “Low Fitness” cohort ($n=74$). The resulting variable indicates whether or not a student met the physical activity recommendations of the “High Fitness” cohort, and this relationship was modeled using a logistic regression.

Each model was additionally fit with demographic covariates, including grade, sex, race, and socioeconomic status, to examine whether the relationship between assessment scores and physical activity persisted after the adjustment.

3.5.1 Evaluation of Alternative Models

PA Days per Week has often been modeled using Poisson regression (Borges et al. 2015; Slymen et al. 2006). Those cases were typically zero-inflated, with observations taking values near the maximum being exceedingly rare. The mean and variance of PA Days, as well as the shape of the distribution, are not well approximated by a Poisson distribution, and thus the Poisson regression model was rejected.

Due to the nested structure of the data, a mixed-effects model was considered. It stands to reason that students that belong to the same classroom and the same school should not be viewed as independent observations. A preliminary mixed-effects model was fitted, and yielded an intra-class correlation coefficient of 0.026. This indicated that a negligible amount of the variation in physical activity was attributable to the classroom/school grouping. A visual inspection of physical activity distributions by classroom and school (see Figure 3.1) also does not show any discernible pattern, and thus a mixed-effects model was rejected.

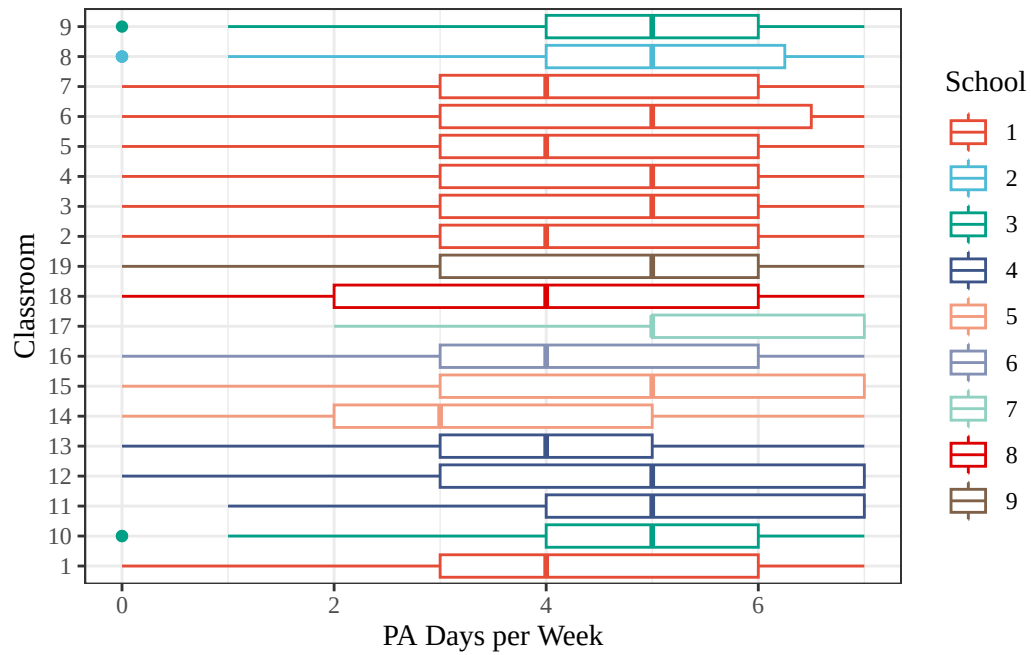


Figure 3.1: Distribution of PA Days by Classroom and School

Results

4.1 Sample Characteristics and Demographics

The sample consisted of 1504 students across 19 classrooms in 9 Indiana secondary schools. Respondents were primarily students early in their secondary education, with the vast majority of respondents being ninth-grade students (84%; $n = 1265$), with smaller cohorts of grades 10 ($n = 82$), 11 ($n = 16$), and 12 ($n = 21$) also included. Grade level was not reported by 120 students. Correspondingly, the age distribution was centered at age 14 ($n = 976$), followed by 15 ($n = 299$), 13 ($n = 157$), 16 ($n = 43$), 17 ($n = 22$) and 18 ($n = 7$).

Respondents were distributed evenly between sexes, with 765 reporting female and 747 reporting male. The majority of respondents were White ($n = 1011$), followed by Black ($n = 237$), Asian ($n = 49$), Native American ($n = 33$), and Native Hawaiian ($n = 8$). Due to small sample sizes, Asian, Native American, and Native Hawaiian respondents were combined into the “Other Race” category. 166 respondents did not report race. Notably, 64% of respondents come from schools associated with higher socioeconomic status, either belonging to the “low poverty” or “mid-low” poverty group, which may limit the generalizability of any findings. The “mid-high” poverty group contained significantly higher proportions of students who reported their race as black or unknown.

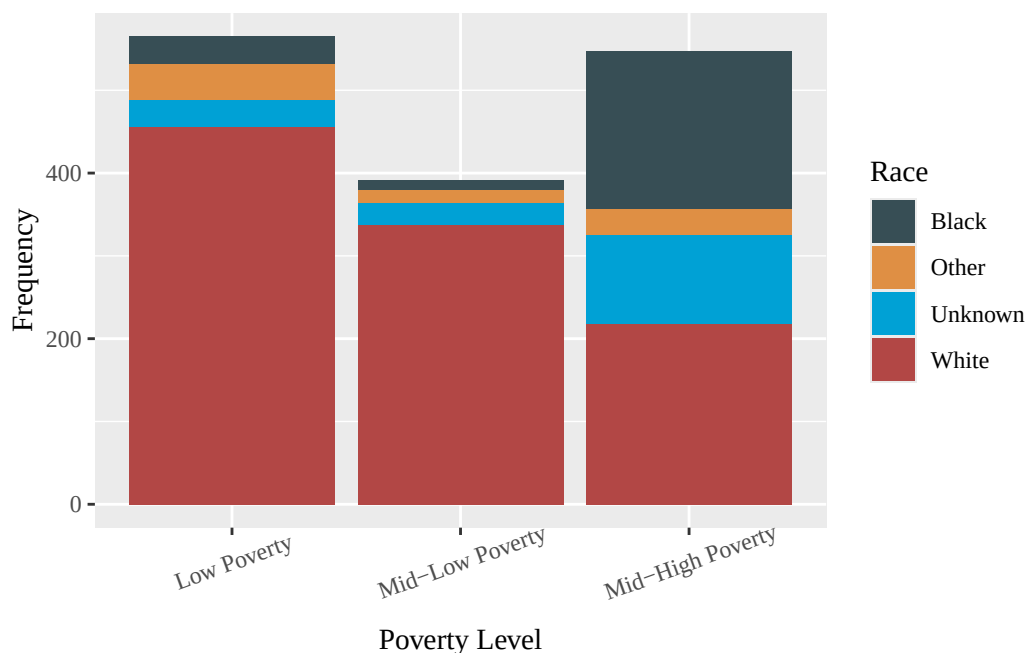


Figure 4.1: Race Composition by Poverty Level

4.2 Descriptive Statistics

4.2.1 Distributions of Test Scores

Assessment scores varied across standards. Standards 2 and 3 were approximately normally distributed, while Standard 4 was significantly skewed left, with most students scoring in the upper range (see Figure

4.2). Standard 4 had the highest median score (71.4%), followed by the overall score (61.5%), Standard 2 (57.1%), and Standard 3 (50%) (see Table 4.1). Visual inspection of distribution highlights the disparity in number of assessment items across standards (21 items for Standard 2, 4 for Standard 3, 14 for Standard 4), which affects the resolution and potentially the reliability of each distribution.

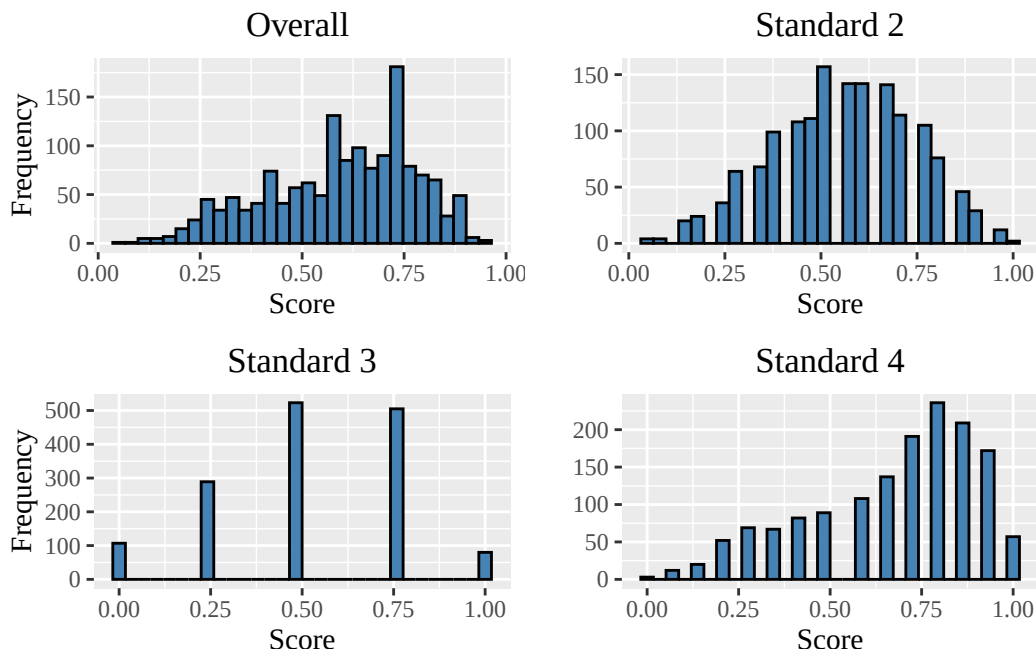


Figure 4.2: Distribution of Test Scores by Standard

Table 4.1: Summary Statistics for Test Scores

Standard	Mean	Q1	Median	Q3
Overall	0.595	0.487	0.615	0.744
Standard 2	0.560	0.429	0.571	0.714
Standard 3	0.527	0.250	0.500	0.750
Standard 4	0.669	0.500	0.714	0.857

4.2.2 Distributions of PA Variables

Students generally reported high levels of physical activity. Figure 4.3 shows the distributions of each physical activity metric. The distribution of PA Days per Week was left-skewed, with a median of 5 days and a most common response of 7 days per week. Similarly, PA Minutes per Day was left skewed, with the most common response being 60+ minutes per day. Strength Days showed a different pattern, with a right-skewed distribution, a median of 3 days, and a most common response of 0 days.

The majority of students fell into the “Medium Fitness” cohort ($n = 1054$). A substantial number of students belonged to the “High Fitness” cohort ($n = 376$), though only a small number of students belonged to the “Low Fitness” cohort ($n = 74$). The low sample size of the “Low Fitness” category limits the statistical power of analyses involving this group.

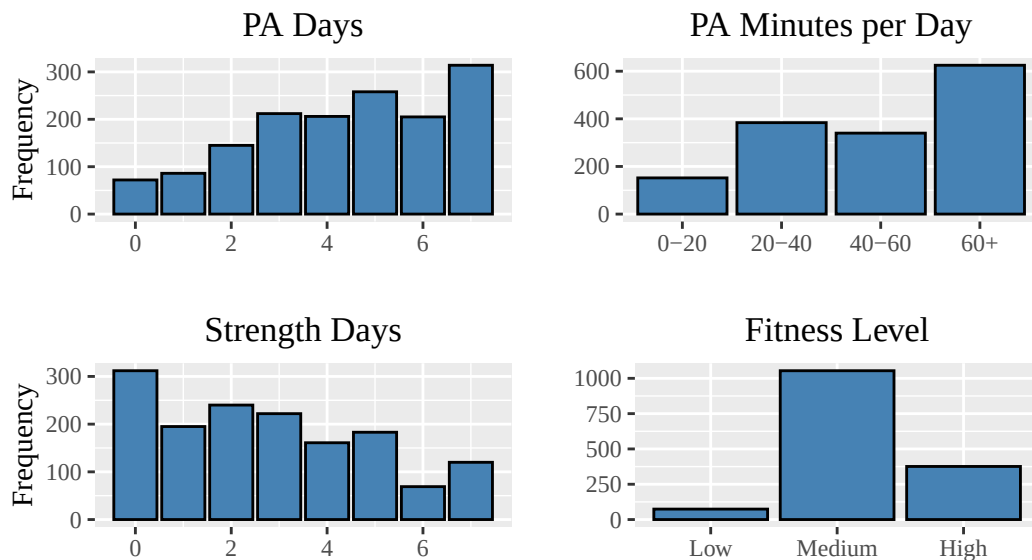


Figure 4.3: Physical Activity Metric Distributions

4.2.3 PA Days and PA Minutes per Day Inconsistency

Students generally reported high levels of physical activity. The median reported days of physical activity was 5, and most students reported 40+ minutes per day of physical activity. There was, however, a notable discrepancy between these statistics. A student who reports an average of 60+ minutes of physical activity per day should also report 7 days of physical activity per week, by definition. While students did report higher average minutes of daily physical activity as they reported higher days of physical activity per week, there was inconsistency in how these metrics were reported. For example, 19 students reported both 0 days of physical activity per week, and an average of 60+ minutes of physical activity per day. The plot of the distribution of PA Minutes per Day by reported PA Days per Week (Figure 4.4) makes this inconsistency clear.

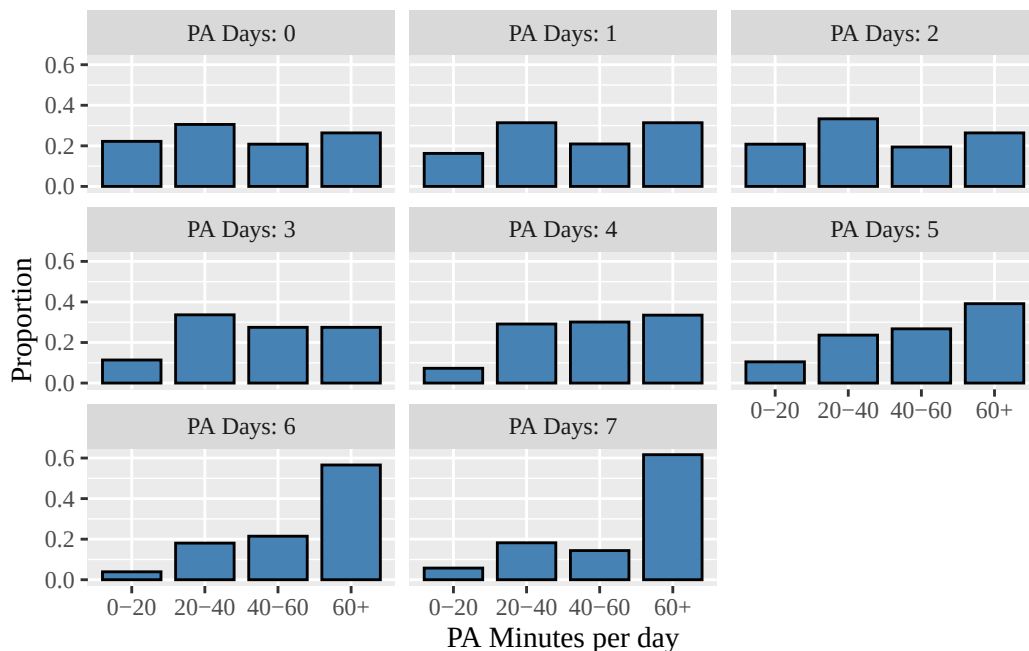


Figure 4.4: Proportional PA Minutes Distribution by PA Days

4.3 Exploratory Findings

4.3.1 Correlation Between Scores and Outcome Variables

Assessment scores generally correlated positively with physical activity across all three outcome metrics, as shown by Figure 4.5. Students in the High Fitness cohort scored higher than the Medium Fitness cohort, which scored higher than the Low Fitness cohort across most standards. The one exception being Standard 3, where the Low Fitness cohort outperformed Medium Fitness. The small sample size of the Low Fitness cohort ($n = 74$) should be considered when interpreting this result.

A similar positive relationship was observed between test scores and PA Days per Week. Students who reported higher numbers of physically active days per week also tended to score higher across each standard. Unexpectedly, there is a marked decline in scores between students who report 6 and 7 PA Days per Week.

The relationship between PA Minutes per Day and test scores was also positive across each standard. There is a notable jump in scores between students who report less than 20 minutes per day and students who report 20-40 minutes, with scores remaining stable across the higher activity categories. This pattern motivated the dichotomization of this variable, as mentioned in the Methods section.

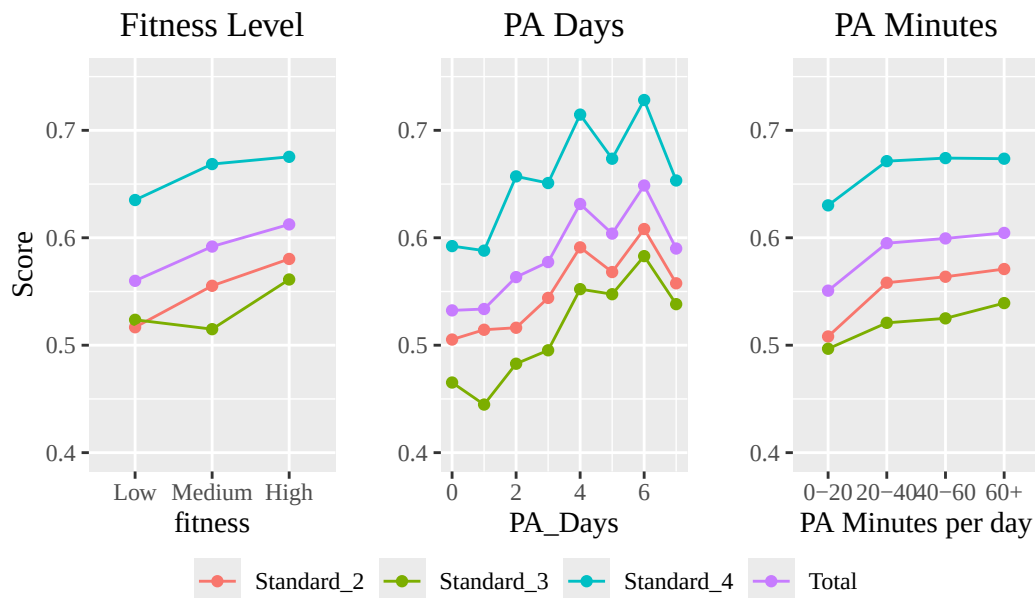


Figure 4.5: Correlation Between Scores and PA Metrics

4.3.2 PE days and PA Days

To examine whether the number of PE Days per Week was associated with activity levels, we compared the distributions of PE Days across each level of PA Days per Week. Students tend to report either 3 or 5 days per week of PE classes, with no discernible pattern across reported days of physical activity. While a chi-square test indicated a statistically significant association ($\chi^2 = 76.57, p < 0.001$), the effect size was negligible (Cramér's $V = 0.101$). This suggests that the raw number of days of PE per week has little impact on the number of days that a student reports being physically active.

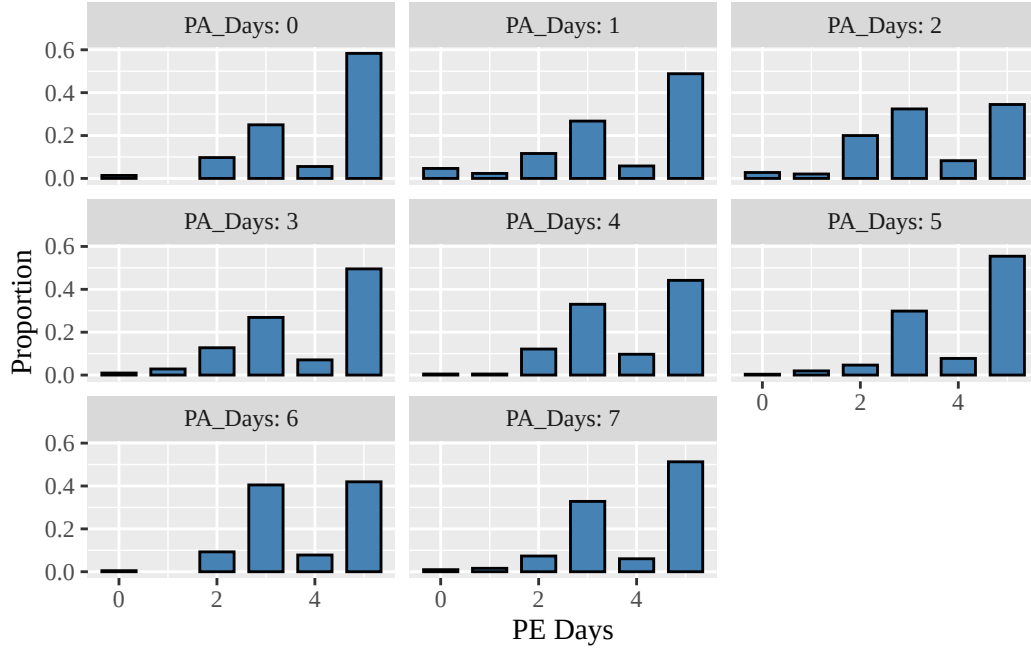


Figure 4.6: Distribution of PE Days by PA Days

4.4 Model Results

4.4.1 PA Minutes Threshold

A logistic regression was fit to examine the association between the overall standardized assessment scores and the probability of a student exceeding the sedentary threshold, as measured by PA Minutes per Day (Table 4.2). The standardized, overall assessment score was statistically significant ($p = 0.001$), which indicates that higher scores are associated with larger odds of a student exceeding the 20-minute sedentary threshold. However, this model offers little explanatory power, with an AUC of 0.586, and McFadden's R^2 of 0.01. No overdispersion was detected in this model. According to this model, an increase of one standard deviation of the overall score is associated with a 1.30-fold increase, or a 30% increase, in the odds of exceeding the 20 minute sedentary threshold. The corresponding 95% confidence interval for the odds ratio is [1.11,1.54].

When standardized assessment scores separated by standard were modeled against the same sedentary threshold variable, only Standard 2 had a significant predictor ($p = 0.004$). Standards 3 and 4 were not significant predictors ($p = 0.893, p = 0.634$, respectively). Model fit remained poor with McFadden's $R^2 = 0.014$ and an AUC of 0.590, consistent with the overall score model. According to this model, an increase of one standard deviation of the Standard 2 score corresponds to a 1.41-fold increase in the odds. The corresponding 95% confidence interval for the odds ratio is [1.12, 1.79].

When demographic covariates were added, the overall score was only marginally significant ($p < 0.1$) and decreased in magnitude. Standard 2 assessment scores remained significant at the $p < .05$ threshold. Among the demographic variables, Sex was marginally significant ($p < 0.1$) and Poverty Level was highly significant ($p < 0.01$). Males were associated with higher odds of surpassing the sedentary threshold, and belonging to the "Mid-High Poverty" class was negatively associated with active behavior, compared to the baseline "Low Poverty" class. Model fit remained modest across all four models, with McFadden's R^2 increasing to 0.034 and AUC to 0.643 with the addition of demographic variables. According to this model, which includes covariates and scores separated by standard, an increase in one standard deviation of the Standard 2 score

corresponds to a 1.35-fold increase in the odds. The corresponding 95% confidence interval for the odds ratio is [1.06, 1.72].

4.4.2 PA Days per Week

As overdispersion was detected in binomial models, a quasi-binomial regression model was fit to examine the association between the same assessment scores and physical activity, this time measured by self-reported physically active days per week (Table 4.3). When fit as the sole predictor, overall score was highly significant ($p < 0.01$). Scoring higher overall on the knowledge-based PE exam is associated with higher odds that a student will be active on a given day. According to this model, an increase by one standard deviation in the overall score is associated with a 1.16-fold, or a 16% increase, in the odds that a student was physically active on any given day of the week. The associated 95% confidence interval for the odds ratio is [1.09, 1.24].

When evaluating the association between individual standards and PA Days, both Standard 2 and Standard 3 were significant ($p < 0.05$, $p < 0.01$, respectively). Standard 4 was not significant, and its coefficient was slightly negative. The magnitude of the association of Standard 3 was slightly larger than that of Standard 2. According to this model, an increase by one standard deviation in a student's score on Standard 3 is associated with a 1.13-fold increase in the odds that a student was physically active on any given day of the week. The corresponding 95% confidence interval of the odds ratio is [1.05, 1.22].

When demographic variables were added, overall scores remained a significant predictor, as well as Standard 2 and Standard 3 assessment scores. According to the model that includes covariates and scores by standard, an increase by one standard deviation in a student's score for Standard 2 and Standard 3 are associated with a 1.12-fold and a 1.09-fold increase in the odds that a student is physically active on a given day, respectively. The associated 95% confidence intervals for the odds ratios for Standard 2 and Standard 3 are [1.02, 1.22] and [1.01, 1.18], respectively. Sex, poverty level, and whether or not a student was Latino were all significant predictors. Males were associated with higher odds of physical activity, and Latino students were associated with lower odds of physical activity. Notably, belonging to the "Mid-Low Poverty" class was associated with higher odds of physical activity, higher than both the "Mid-High Poverty" and "Low Poverty" classes.

4.4.3 Fitness Cohorts

A logistic regression was fit to model the relationship between assessment scores and the probability of students meeting physical activity recommendations, and thus belonging to the "High Fitness" cohort (Table 4.4). When fit as the only predictor, overall score had a significant ($p < 0.05$) positive relationship with the probability that a student belongs to the "High Fitness" cohort. The model offers poor explanatory power (McFadden's $R^2 = 0.003$, $AUC = 0.542$). According to this model, an increase by one standard deviation in a student's overall score is associated with a 1.14-fold (14%) increase in the odds that the student is in the "High Fitness" cohort. The associated 95% confidence interval is [1.01, 1.28].

With each standard included separately, the model is similarly poor-fitting (McFadden's $R^2 = 0.009$, $AUC = 0.570$). Standard 2 and Standard 3 had significant ($p < 0.05$, $p < 0.01$, respectively) positive associations with the probability of belonging to the "High Fitness" cohort. Standard 4 is also significant ($p < 0.05$), but the relationship is negative. This result is surprising, but should be viewed with the context of poor overall model fit.

When demographic variables are added, the resulting models had substantially improved fits (McFadden's $R^2 = 0.039$, $AUC = 0.635$), consistent with expectations. Overall score had its significance level attenuated ($p < 0.1$). Standard 2 maintained the same level of significance ($p < 0.05$), Standard 3 saw its significance attenuated ($p < 0.1$), and while Standard 4 maintained a negative coefficient, the relationship is not significant. According to the model that includes covariates and scores by standard, an increase by one standard deviation in a student's score on Standard 2 is associated with a 1.23-fold increase in the odds that a student belongs to the "High Fitness" cohort. The associated 95% confidence interval for the odds ratio is [1.04, 1.46].

Table 4.2: Logistic Regression: PA Minutes Threshold

	<i>Dependent variable:</i>			
			Active Daily (>20 min)	
	Overall	Standards	Overall and Covariates	Standards and Covariates
	(1)	(2)	(3)	(4)
Overall Score	0.267*** (0.083)		0.171* (0.093)	
Standard 2		0.346*** (0.120)		0.299** (0.123)
Standard 3		0.013 (0.098)		−0.082 (0.103)
Standard 4		−0.058 (0.121)		−0.061 (0.124)
Sex			0.342* (0.177)	0.339* (0.177)
Mid-Low Poverty			−0.039 (0.255)	−0.023 (0.256)
Mid-High Poverty			−0.732*** (0.237)	−0.749*** (0.237)
Other Race			0.016 (0.420)	0.046 (0.422)
Unknown Race			0.078 (0.317)	0.086 (0.318)
White			0.084 (0.250)	0.107 (0.251)
Latino			0.138 (0.272)	0.127 (0.273)
Constant	2.212*** (0.088)	2.223*** (0.089)	2.288*** (0.285)	2.288*** (0.286)
McFadden's R ²	0.010	0.014	0.031	0.035
AUC	0.586	0.590	0.639	0.643
Observations	1,501	1,501	1,495	1,495
Log Likelihood	−486.983	−485.356	−476.129	−474.148
Akaike Inf. Crit.	977.966	978.711	970.258	970.297

*Note:**p<0.1; **p<0.05; ***p<0.01
Standard errors in parentheses.

Table 4.3: Binomial Regression: PA Days per Week

	<i>Dependent variable:</i>			
	PA Days per Week (0-7)			
	Overall	Standards	Overall and Covariates	Standards and Covariates
	(1)	(2)	(3)	(4)
Overall Score	0.150*** (0.032)		0.152*** (0.036)	
Standard 2		0.105** (0.045)		0.113** (0.045)
Standard 3		0.125*** (0.038)		0.090** (0.039)
Standard 4		-0.024 (0.047)		-0.003 (0.047)
Sex			0.510*** (0.066)	0.502*** (0.066)
Mid-Low Poverty			0.338*** (0.085)	0.329*** (0.086)
Mid-High Poverty			0.036 (0.088)	0.050 (0.088)
Other Race			-0.158 (0.160)	-0.169 (0.161)
Unknown Race			-0.111 (0.129)	-0.126 (0.129)
White			0.101 (0.103)	0.091 (0.103)
Latino			-0.308*** (0.104)	-0.311*** (0.104)
Constant	0.513*** (0.032)	0.514*** (0.033)	0.162 (0.110)	0.173 (0.111)
Observations	1,498	1,498	1,495	1,495

*Note:**p<0.1; **p<0.05; ***p<0.01
Standard errors in parentheses.

Table 4.4: Logistic Regression: Fitness Cohort

	<i>Dependent variable:</i>			
	Meets Fitness Recommendations			
	Overall	Standards	Overall and Covariates	Standards and Covariates
	(1)	(2)	(3)	(4)
Overall Score	0.128** (0.061)		0.132* (0.068)	
Standard 2		0.205** (0.083)		0.210** (0.086)
Standard 3		0.198*** (0.071)		0.122* (0.073)
Standard 4		-0.198** (0.087)		-0.146 (0.090)
Sex			0.647*** (0.126)	0.628*** (0.126)
Mid-Low Poverty			0.551*** (0.149)	0.534*** (0.150)
Mid-High Poverty			-0.397** (0.175)	-0.375** (0.176)
Other Race			-0.426 (0.330)	-0.449 (0.331)
Unknown Race			-0.067 (0.259)	-0.089 (0.259)
White			-0.195 (0.206)	-0.207 (0.207)
Latino			0.167 (0.203)	0.162 (0.204)
Constant	-1.098*** (0.060)	-1.109*** (0.060)	-1.327*** (0.221)	-1.313*** (0.222)
McFadden's R ²	0.003	0.009	0.036	0.039
AUC	0.542	0.570	0.624	0.635
Observations	1,499	1,499	1,494	1,494
Log Likelihood	-842.062	-836.134	-811.535	-808.435
Akaike Inf. Crit.	1,688.123	1,680.269	1,641.070	1,638.869

*Note:**p<0.1; **p<0.05; ***p<0.01
Standard errors in parentheses.

Discussion

The purpose of this study was to demonstrate the potential of standards-based survey data to identify associations between PE knowledge and self-reported physical activity behavior among high-school students. Leveraging a dataset that included SHAPE standards-based assessment items and YRBS self-reported behavior metrics, we investigated the relationship between standards-based PE knowledge scores and three physical behavior metrics: PA Minutes per Day, PA Days per Week, and Fitness Cohort. Across all three measures, increased PE knowledge scores were associated with increased probabilities of higher levels of physical activity, even after controlling for demographic variables. While the effect sizes of these findings were small, these findings support the broader argument that PE knowledge may play a meaningful role in promoting positive physical behaviors in students.

Across the three outcome measures, Standard 2, which assesses students' knowledge of fitness concepts, was the most consistent predictor of physical activity behavior out of the three knowledge-based standards. This suggests that students who have a stronger grasp of fitness concepts are more likely to engage in physical activity, even when controlling for their demographic background. From the standpoint of curriculum design, and given the weak explanatory power of these models, this would tentatively support the prioritization of teaching fitness concepts over the other PE knowledge-based standards.

Standard 3 had a less straightforward relationship with the three outcome measures. It was a significant predictor of physical behavior in the PA Days model and the unadjusted fitness cohort model, but was not significant in the adjusted fitness cohort model or the PA Minutes models. Since Standard 3 assesses the social component of physical activity in students, it's possible that students whose social lives involve physical activity may be more likely to engage in physical activity more often, but not necessarily at a higher intensity.

Standard 4, which assesses the degree to which personal identity relates to movement, had either insignificant or significantly negative relationships across the three outcome variables. This could be interpreted as a genuine disconnect between personal identity around physical activity and their actual behavior, or it could be seen as an unreliable result of models with weak explanatory power.

Demographic variables, particularly sex and poverty level, were consistently significant predictors of physical activity across all three outcome measures, more than the knowledge-based standards. Male students were consistently associated with higher levels of physical activity across every model.

Poverty level showed an interesting but somewhat unexpected pattern. Students belonging to the "Mid-High Poverty" class were consistently associated with lower levels of physical activity, while the "Mid-Low Poverty" class was associated with higher activity levels than even the "Low Poverty" class in the fitness cohort and binomial models. It's difficult to draw conclusions based on this unexpected result, and further investigation with a more representative sample would be needed to assess whether this is a meaningful relationship or an artifact of the data.

Latino students were negatively associated with physical activity in the binomial model, but that relationship did not reach significance in the other models. Race generally did not reach significance in the models when controlling for other demographic variables. The limited racial diversity of the sample limits the conclusions that can be drawn about the relationship between race and physical activity.

There are several important limitations of this study that should be considered when interpreting these findings. First, model fit was consistently poor across all outcome measures, with low McFadden's R^2 and AUC values. While variables still reached significance, SHAPE knowledge standards explain only a small portion of the variance in physical activity behavior. There are likely many other factors that influence students' physical behavior that are not captured in this dataset.

Second, all physical activity metrics were based on self-report, which are subject to response bias. The inconsistency between PA Days and PA Minutes, noted in the Results section, demonstrates that students may not be interpreting or responding to these questions consistently.

Third, the sample is not representative of the broader population of US students. The vast majority of students were white, from a higher socioeconomic class, and located in a single state (Indiana). This limits the generalizability of any findings.

Fourth, the knowledge assessment items were originally developed under the 2004 SHAPE Standards framework, then mapped onto the appropriate umbrella standard under the current four-standard framework. This mapping may introduce some uncertainty into the interpretation of scores by standard.

The findings of this study, while modest, demonstrate the potential of standards-based survey data to meaningfully answer questions about the relationship between PE knowledge and physical activity behavior. This study should be viewed as a proof-of-concept to motivate further research, rather than as presenting definitive results.

A more representative study, that includes broader participation among socioeconomic class, race, and location, would allow for more generalizable conclusions about the relationship between PE knowledge and physical activity behavior.

Future research would also benefit from more reliable outcome measures. The inconsistency between self-reported PA Days and PA Minutes demonstrates the unreliability of students' reporting of these metrics, and a future study would benefit from taking steps to mitigate these inconsistencies at the time of survey. Further, an assessment based on the current four-standard framework would strengthen the results of the study.

The goal of this line of research is to provide PE educators with an empirical framework for curriculum design. If larger-scale studies are able to confirm some of the results in this study while addressing the previously-mentioned limitations, such research could serve to help design more standardized, evidence-based PE programming to promote better health outcomes for students.

References

- Borges, A., T. N. Gomes, D. Santos, et al. 2015. “A Count Model to Study the Correlates of 60 Min of Daily Physical Activity in Portuguese Children.” *International Journal of Environmental Research and Public Health* 12 (3): 2557–73. <https://doi.org/10.3390/ijerph120302557>.
- Lund, J., and D. Tannehill. 2015. *Standards-Based Physical Education Curriculum Development*. 3rd ed. Jones & Bartlett Learning.
- Pfledderer, C. D., E. J. Mullane, D. M. Y. Brown, et al. 2025. “Five-Year Trends in u. S. Child and Adolescent 24-Hour Movement Behavior Guideline Adherence, 2018–2022.” *Journal of Activity, Sedentary and Sleep Behaviors* 4 (1): Article 16. <https://doi.org/10.1186/s44167-025-00086-8>.
- Richards, K. A. R., T. J. Templin, and K. C. Graber. 2014. “The Socialization of Teachers in Physical Education: Review and Recommendations for Future Works.” *Kinesiology Review* 3 (2): 113–34. <https://doi.org/10.1123/kr.2013-0006>.
- SHAPE America – Society of Health and Physical Educators. 2024. “National Physical Education Standards.” <https://www.shapeamerica.org/standards/pe/>.
- Slymen, Donald J., Guadalupe X. Ayala, Elva M. Arredondo, and John P. Elder. 2006. “A Demonstration of Modeling Count Data with an Application to Physical Activity.” *Epidemiologic Perspectives & Innovations* 3 (1): 3. <https://doi.org/10.1186/1742-5573-3-3>.
- Tremblay, M. S., A. G. LeBlanc, M. E. Kho, et al. 2011. “Systematic Review of Sedentary Behaviour and Health Indicators in School-Aged Children and Youth.” *International Journal of Behavioral Nutrition and Physical Activity* 8: Article 98. <https://doi.org/10.1186/1479-5868-8-98>.
- Underwood, J. M., N. Brener, J. Thornton, et al. 2020. “Overview and Methods for the Youth Risk Behavior Surveillance System — United States, 2019.” *MMWR Supplements* 69 (1): 1–10. <https://doi.org/10.15585/mmwr.su6901a1>.

Appendix

6.1 Appendix A: PE Knowledge Assessment Items

The following 39 items comprise the PE knowledge assessment, organized by SHAPE National Standard.

6.1.1 Standard 2

S2 Analyzes physical development across the lifespan that influences physical activity choices

S2PT1. Which of the following is the most accurate statement about physical activity levels? A. If children are not physically active, they won't be physically active when they are adults. B. Physical activity patterns stay the same as people age. C. Children and older adults are not that physically active. D. Physical activity patterns change as people age.

S2PT2. Tommy's grandmother played basketball while growing up and still rides a bike and walks regularly. All of the following are benefits that Tommy's grandmother could realize from physical activity, except: A. She sleeps better. B. She strengthened her bones to help resist osteoporosis. C. She has sore muscles. D. She has stronger heart and lungs.

S2PT3. Mohammad is ahead of his peers in terms of motor skill development, which may be because of more: A. Maturity and practice. B. Instruction and speed. C. Practice, instruction and body size. D. Instruction, practice and maturity.

S2PT4. As you continue to mature, which of the following is most likely to happen? A. I will get faster and stronger as I grow older, and that affects my skills. B. My skills will stay the same if I continue to practice. C. My skills in most activities will increase if I practice. D. My skills will be better immediately.

6.1.2 Standard 3

S3 Implements a personal physical activity plan that is part of a healthful lifestyle

S2PT5. A pedometer is usually used to record how physically active you are by: • Recording your peddling action while you ride your bike. • Measuring your average heart rate while you walk. • Counting the number of steps you take during physical activity. • Recording the number of physical activities in which you participate weekly.

S3PT6. You should choose activities for your personal physical activity plan that: • You can fit into your schedule.

• You can perform with enjoyment. • Your friends enjoy doing. • Are provided by your local recreation program.

Employs self-management skills to maintain a physically active life

S3PT7. Which of the following are benefits of being physically fit? • Decreased heart rate and appetite. • Improved self-esteem and reduced stress. • Increased resting heart rate and lowered stress. • More energy and less need for sleep.

S3PT8. A physically active person accumulates up to 60 minutes a day of moderate-to- vigorous physical activity. Which of the following best describes a "physically active lifestyle?" • Walks to school, mows grass and rakes leaves, cleans house, completes a personal fitness workout 5 to 6 days a week, plays on the school tennis team. • Rides the bus to school, participates in PE class 2 days a week, works at the local grocery store stocking shelves on weekends. • Completes homework each day, completes chores at home, practices

daily with the marching band at school, walks to school. • Participates in a bowling league 1 to 2 days a week, volunteers at the community library restocking books 2 to 3 days a week, runs a mile 3 days a week.

S3PT9. Which of the following plans demonstrates the most physically active lifestyle? • Work out at the local fitness center at least 3 days a week. • Swim or play soccer or tennis for 60 minutes at least 5 days a week. • Run a mile at least 2 days a week. • Perform a weight-lifting program at least 3 days a week.

S3PT10. Which group of activities would most likely help to improve all components of health-related physical fitness if added to your personal physical activity plan? • Biking, volleyball, swimming. • Soccer, basketball, softball. • Gymnastics, running, bowling. • Yoga, weight lifting, running.

Achieves levels of health-related criterion referenced standards for fitness.

S3PT11. The percentages of fat, muscle, bone and water in the body are fitness components called:

- Weight management. • Nutritional balance. • Body composition. • Physical structure.

S3PT12. Running for distance or time can be used to measure: • Flexibility. • Aerobic endurance. • Distance capacity. • Muscle strength.

Demonstrates the skill, knowledge and desire to monitor and adjust activity levels to meet personal fitness needs.

S3PT13. You can increase the intensity of your workout by:

- Increasing distance & speed. • Decreasing speed & distance. • Increasing time & decreasing distance. • Increasing time & decreasing repetitions.

S3PT14. In the FITT guidelines, “frequency” refers to: • How long you exercise. • How much time you allow between repetitions. • How often you exercise. • The number of weeks in a workout plan.

S3PT15. When you reach your initial goal for improved physical fitness, you should apply the principle of “progression” to safely adjust your work-out plan to continue fitness improvement by: • Increasing the intensity of your workout. • Increasing the number of “no-workout” days per week. • Keeping your workout the same. • Work out at a lower level of intensity.

S3PT16. The type of activities to improve aerobic endurance must be able to _____ throughout the activity. • Keep the heart rate at the target heart rate level. • Keep the heart rate at a resting level. • Change the heart rate. • Allow the heart rate to recover.

S3PT17. My goal is to lose weight safely and decrease my percentage of body fat in the next four weeks by _____. • Walking daily, reducing water consumption, eating more whole grain bread and eliminating fatty foods. • Walking 30 minutes daily, eating more fruit and vegetables, and consuming more water and fewer calories. • Jogging every other day, avoiding meat and bread, eating more fruit, and drinking 8 glasses of water each day. • Walking 3 times per week, cutting my calorie intake by 25 percent, eating more protein, and increasing my water intake daily.

S3PT18. My goal is to complete the mile run without stopping to rest by doing the following for four weeks:

- Alternately run and walk at least 5 days a week, increasing the distance I run gradually. • Walk briskly for 1¼ miles at least 5 days a week. • Alternately run and walk at least every other day. • Walk 1 mile briskly 3 days the first week and add ½ mile each week after that.

S3PT19. When planning your exercise program, you should start by: • Using a plan of physical fitness activities based on nationally set goals. • Keeping a log of your participation in all physical activities. • Reviewing your personal fitness test scores and setting personal fitness goals. • Deciding how much time you have to exercise.

S3PT20. When the length of a run is changed from 1 mile to 1¼ miles, what principle is being applied? • Intensity. • Specificity. • Frequency. • Progression.

S3PT21. When adjusting your workout to increase flexibility, stretch each muscle group beyond the distance required by your normal activities and until: • You cannot stand the pain. • The muscle becomes tight without painful discomfort. • You reach the recommended distance for your age group. • You reach your long-term goal distance.

S3PT22. A personal fitness workout plan for improving total health-related fitness should include goals based on:

- Personal evaluation, with specific activities to address body composition.
- Published charts of expectations for a person of your height and weight, with specific activities to address all fitness components.
- Personal test results, specific activities to address each component, and realistic expectations.
- Personal test results and desired achievement of specific sports skills.

Scenario: Goal-Setting. Use to answer questions 21 & 22.

Jamal wants to set up a fitness plan that will improve his fitness levels. He is 15 years old and is in high school. He appears to be healthy but is overweight. Below are his pre-test scores and goals for his fitness plan. Read the goals and answer the questions below.

Jamal's Goals for September through January 31.

Fitness Component Pre-Test Goal Muscle endurance Push-ups = 13

Curl-ups = 12 20+ Body composition 32% body fat 32% body fat or less Aerobic endurance 22-minute mile 18-minute mile

S3PT23. Jamal has set a goal for improving his push-up score. Which of the following work-out plans is most likely to help him reach his goal?

- Complete 3 sets of up to 10 modified push-ups each, with hands on a table and feet on the floor, every other day for 3 weeks.
- Complete 3 sets of up to 10 modified push-ups each, with hands on a table and feet on the floor, at least 2 days a week for 3 weeks.
- Complete 3 sets of up to 10 modified push-ups each, with hands on a table and feet on the floor, at least once a week for 3 weeks.
- Complete as many push-ups as you can 3 days a week for 3 weeks.

S3PT24. Jamal has set a goal of improving his curl-up score by completing 3 sets of up to 10 curl-ups each, every other day for 3 weeks. If he reaches his goal of 20 curl-ups without rest, Jamal should adjust the goal and continue the workout for 3 more weeks, but:

- Increase the number of repetitions to up to 15 in each set.
- Keep the number of repetitions in each set the same.
- Decrease the number of repetitions and increase the number of sets in each workout.
- Complete as many curl-ups as possible in each workout.

S3PT25. When designing a conditioning program to improve health-related fitness, you should first look at your current fitness test scores to determine realistic goals around which to create a workout plan. Which set of guidelines must you consider for each of the activities/exercises in the daily plan?

- Frequency, intensity, time and specific type of exercise needed to address each health-related fitness component.
- Personal needs for sleeping, studying, working and eating.
- Your physical activity level and interests.
- Nutrition needs and eating habits.

Applies problem-solving techniques.

S3PT26. If Matt has the solution to a problem that his group is trying to answer, he should:

A. Tell the group the answer quickly so that it can be implemented quickly.

B. Become the group leader and guide group members to complete the task using his solution.

C. Present his idea to the group and ask for alternative ideas.

D. Task group members with solving the problem consistent with his solution.

S3PT27. How should Samantha act when the official makes a call with which she disagrees?

A. Let everyone know that it was a bad call.

B. Accept the official's call and play by the spirit of the game.

C. Point out the mistake to the official.

D. Wait until after the game and then tell the official that it was a bad call.

Demonstrates responsibility for following safe practices, rules, procedures and etiquette in physical activity settings.

S3PT28. What should you do if an opponent drops his/her racket accidentally during a competitive net game and he/she cannot play the point?

A. Offer to replay the point.

B. Continue to play.

C. Stop play and give yourself a point.

D. Declare that dropping a racket is a safety violation, which makes you the winner.

Exhibits supportive behavior.

S3PT29. Good sportsmanship means that, when you win a game, you should:

- A. Not say or do anything.
- B. Thank the opposing team for a good game.
- C. Congratulate your teammates.
- D. Celebrate your win enthusiastically

6.1.3 Standard 4

Recognizes the different benefits associated with participation in various types of physical activities

S4PT30. Of the following activities that Kaitlin enjoys regularly, which should she select if she wants the most fitness benefit? A. Lap swimming. B. Tai chi. C. Walking. D. Volleyball.

S4PT31. Charice wants to increase her participation in physical activity so that she is active every day. What criteria should she use for selecting an activity? A. Select the activity that her friends say they like to do. B. Select an activity that she doesn't know how to do. C. Select an activity that is enjoyable to her. D. Select the most popular activity among classmates.

S4PT32. Which would best serve to help Carla improve or maintain her flexibility? A. Weight training. B. Curl-ups. C. Line dancing. D. Gymnastics.

S4PT33. Devin's doctor has warned him of the risk factors associated with his obesity and has recommended that he improve his cardiorespiratory fitness. Devin should begin by: A. Taking a brisk evening walk every day. B. Jogging every day for 30 minutes. C. Completing 3 to 4 sprints down the football field 3 times a week. D. Jogging three times a week for 30 minutes.

S4PT34. Which one of the following lifestyle changes to improve physical activity doesn't require much effort and could improve your health? A. Taking the stairs instead of the elevator. B. Playing basketball at a local community league. C. Organizing a group to go on a backpacking trip. D. Working out to an exercise video every morning before going to school.

Recognizes the attraction of participation in various physical activities.

S4PT35. Tara's favorite physical activity is mountain biking. She might enjoy that activity most because: A. She enjoys a challenge and being outdoors. B. It requires inexpensive equipment. C. She can do it anywhere. D. She is not skilled at most other activities.

S4PT36. Your friend is not very active. How can you best encourage him to start exercising? A. Invite him to join you in an enjoyable activity. B. Show him a list of those activities that you believe are fun and challenging. C. Encourage him to seek the advice of athletes. D. Give him a handout from class on the benefits of being active.

S4PT37. Students who are physically active and watch what they eat usually: A. Earn better grades in school. B. Are not well liked by other students. C. Are healthier. D. Don't have a lot of free time.

S4PT38. A good choice of physical activity for a person who wants to be physically active but hates sports that include running would be: A. Jogging. B. Dance. C. Soccer. D. Wii sports.

S4PT39. Other than health benefits, why would Carli value rock climbing, mountain biking and kayaking? A. Social interaction. B. The challenge. C. Self-expression. D. Cooperation.

6.2 Appendix B: Grade and Race Distribution Tables

Table 6.1: Grade Distribution

Grade	n	%
9	1265	0.841
10	82	0.055
11	16	0.011
12	21	0.014
Not Reported	120	0.080

Table 6.2: Race Distribution

Race	n	%
White	1011	0.672
Black	237	0.158
Other	90	0.060
Unknown	166	0.110